24. THE LIVER : GENERALITIES

DURATION : 11:53 STUDY SHEET

COURSE SUMMARY

This video explores the complex embryological origin of the liver, composed of tissues from the upper, middle, and lower intestine, as well as essential mesodermal structures. It details the liver's functional duality, acting both as an exocrine gland for bile secretion and an endocrine gland thanks to its mesodermal component. It also highlights the importance of embryonic growth and haemodynamics in the development of this vital organ. Furthermore, the video emphasises the functional integrity and anatomical relationships between the liver and the diaphragm, illustrates how imbalances can lead to dysfunctions, while stressing the diaphragm's role as the 'gate of life' directly influencing liver health.

THE LIVER: GENERALITIES

1. EMBRYOLOGICAL ORIGIN OF THE LIVER

The **liver** has a complex embryological origin, formed from three parts:

- The foregut
- The midgut
- The hindgut

The junction between the foregut and the midgut is the site of an aspiration phenomenon, known as a metabolic field of **loosing fields**. This aspiration field initiates the development of the liver.

Observing a histological section, we see that this tissue, initially a collection of cells, undergoes an aspiration movement. Below this primordial tissue is a **mesodermal** tissue, which will later include the arterial system of the liver, of vitelline origin.

This process of aspiration and fractionation is common to the development of all glands. Each small budding recreates a new aspiration field, thus forming a set of small channels, the **hepatic ducts**.

2. DISTINCTION BETWEEN EXOCRINE AND ENDOCRINE GLANDS

There are two types of glands: **exocrine** and **endocrine**.

- An endocrine gland loses its lumen and secretes its substance directly into the blood.
- The liver, on the other hand, is an exocrine gland. It secretes its biliary substances inside its lumen, returning the sugars necessary for digestion to the digestive system.

In summary, the liver is an aspiration field that associates with the mesoderm, particularly with vessels of vitelline origin, to form its structure. It originates from the intestine but integrates mesodermal components.

3. DUAL ENDODERMAL AND MESODERMAL FUNCTION OF THE LIVER

The liver has a **dual function**:

- **Endodermal function:** of a digestive and specialised cannobiliary type, characterised by the **enterohepatic cycle**, which is very important for the secretion of bile, which is then reabsorbed at the caecal level.
- **Mesodermal function:** an endocrine function, linked to its mesodermal component.

This duality explains how the liver, even after partial ablation, can regenerate its mesodermal structure.

4. ROLE OF THE LIVER IN GROWTH AND HAEMODYNAMICS

For an aspiration field to exist, there must be a phase of **longitudinal growth** of the embryo. This aspiration field is therefore linked to growth, participating in the processes of **cephalisation**, **cardialisation**, and **hepatisation**.

The primary function of the liver is essentially **haemodynamic** and enables a potential for life. Its first impulse is to receive the products of disassimilation, i.e., the catabolic products of the entire body.

This means that the liver is fundamentally designed to participate in **regeneration**. It is the main organ of bodily regeneration, whether at the tendinous, muscular, or other structural levels.

The liver receives multiple inputs: portal, lymphatic, and cardiac.

5. FUNCTIONAL INTEGRITY AND ANATOMICAL RELATIONSHIPS

In osteopathy, the loss of **functional integrity** of the liver with the diaphragm is crucial. The **triangular ligament**, the **falciform ligament**, and the capsule of the liver are in tissue, cellular, and even molecular continuity with the diaphragm and pericardium.

This origin extends beyond the precordial plate. Undifferentiated **mesenchymal** cells, originating from this area, give rise to the **septum transversum** and the pericardium.

The septum transversum, among other things, forms the triangular ligament of the liver, which extends into the hepatic falciform ligament. It also contributes to the **greater omentum**.

The heart, diaphragm, and part of the liver are intimately linked. The liver acts as a fulcrum that indirectly contributes to the formation of the diaphragmatic crura.

The septum transversum, in association with the pericardium, remains stable, while the surrounding structures grow around it, forming this integration through the movement of cerebral development.

A simple imbalance, such as a costal displacement, can affect the position of the diaphragm and the liver.

6. THE DIAPHRAGM: GATEWAY TO LIFE

The diaphragm is often referred to as the **"gateway to life"** due to its orifices and essential role. The interaction with the liver is central.

The rhythmic movement of the diaphragm, with approximately **23,000 breaths per day**, makes the work of the liver dependent on diaphragmatic and pulmonary movement.

We observe differential pressures in the body:

- Negative pressure at the pulmonary level.
- Positive pressure at the abdominal level.
- Negative pressure at the pelvic level.
- Positive pressure in the upper body.

7. IMPORTANCE OF ABDOMINAL NORMOTENSION

Abdominal normotension is an important reference: epigastric tension is slightly higher than hypogastric tension.

- Frequent abdominal hypertension can increase tension in the upper area and decrease strength in the lower areas.
- Treating this area often involves considering the depressive zone.
- Treating the lungs amounts to treating the peritoneum.

The destabilisation of one part leads to the destabilisation of the other. It is therefore crucial to work on the spaces of **volaemia** to support the diaphragmatic process. The liver, being voluminous, and its proximity to the heart and lungs, creates a vital exchange space.

It is necessary to determine the type of abdominal tension and to work on the volaemic patterns. Supporting this area is essential because a loss of function here will decrease overall vitality. The diaphragm will then strive to maintain balance.

8. LIVER, FLUIDS, AND CARDIAC RELATIONSHIPS

The liver manages the qualities of the **fluids** exchanged, particularly for digestion and volume.

It quantifies and modifies the portal tension it transmits to the heart. In reality, the liver is a specialisation of the **duodenum**, which receives a multitude of information (autocrine, paracrine, neurocrine, endocrine).

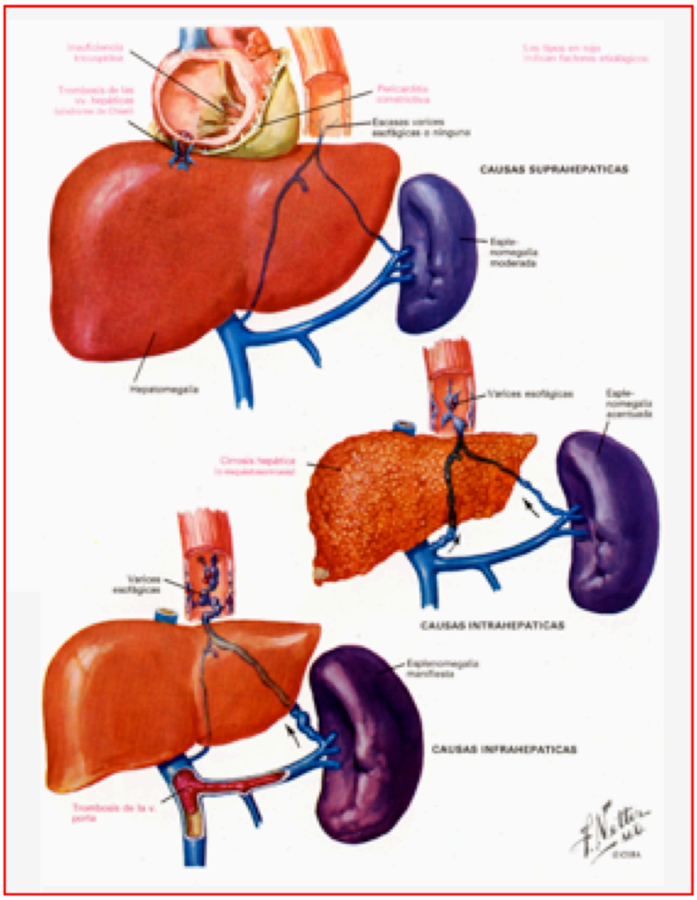

FIG. — Portal hypertension

The duodenum is a crossroads of information for the pancreas and the liver, providing digestive information based on inputs. A good digestive state is therefore paramount.

When the liver loses its movements and the quality of its exchanges, particularly at the lymphatic level, this can destabilise the heart and make it ill. There is a very important relationship between the heart and the liver.

- If the liver begins to destabilise, the heart can become ill.

- Conversely, relieving the liver can help a suffering heart.

Many cardiac affections are actually **metabolic dysfunctions** of the liver.

9. LIVER AND LYMPHATIC SYSTEM

The liver can overload the **lymphatic system**, which in turn overloads the main lymphatic duct and the lateral neurological plexus of the sympathetic chain, leading to distant repercussions.

For example, a painful left shoulder can be a sign of **hepatic congestion**. These are early indicators of hepatic dysregulation, the large primary conduit being first on the left.

One must not forget the **"trinity"** heart-liver-spleen, to which the mesenteric system is added. These "sponges" or states of volaemia work in synergy.

The axis between the liver and the heart is the **vena cava**, whose construction is complex.

10. LIVER AND IMMUNE SYSTEM

There is an important immune relationship involving the **thymic system**. The liver is involved in the passage of **T lymphocytes** with the thymus, and the return of immune information via the spleen. This process contributes to thymic balance and the maturation of T lymphocytes.

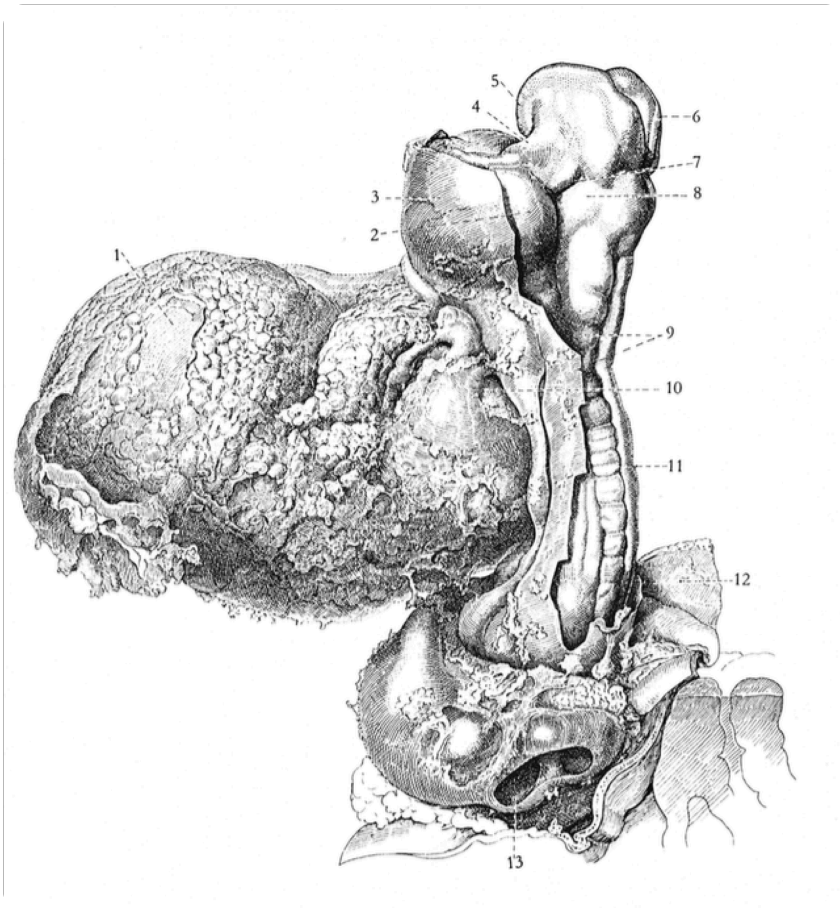

FIG. — Human embryo dissection